



Nehru Skill Development Mission

Established Under Act 1882, Registered Under Govt of Tamilnadu
National Development Organisation, Promoted by Govt of India
Central Board of Examination

DIPLOMA IN X-RAY TECHNOLOGY

TWO YEARS (FIRST YEAR) EXAMINATION

COMMUNICATIVE ENGLISH AND COMPUTER FUNDAMENTALS

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Define Pronoun?
- 2) What is meant by adverb?
- 3) Define active and passive voice?
- 4) Define kinds of verbs?
- 5) Define proper noun?
- 6) Define kinds of adjectives?
- 7) What are the types of Articles?
- 8) Write short notes on 'macro'
- 9) Explain any five functions in MS Excel
- 10) Enter

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Explain few sentences using "shall, will"?
- 2) Explain few sentences using "across, at"?
- 3) Icons
- 4) Explain any five options for the format 'menu'
- 5) Importance of information technology

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Print view
- 2) Techniques for moving an injured (or) ill person.
- 3) Explain in detail about Degrees of comparison?
- 4) Explain any five options for the format 'menu'
- 5) Working areas of computer?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Write a letter to the station master requesting him to reserve two coaches for your school excursion party
- 2) Explain briefly the job descriptions
- 3) Briefly explain storage unit
- 4) Introduction to Pagemaker? Explain



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TWO YEARS (FIRST YEAR) EXAMINATION

AHE003-02 ANATOMY & PHYSIOLOGY

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Heart Chambers
- 2) Bone of the TM joint
- 3) Neurons
- 4) Meninges
- 5) Ovulation
- 6) Pituitary adenoma
- 7) Floating ribs
- 8) Spinal cord
- 9) Orthology
- 10) Voluntary muscles

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Pancreas
- 2) Pelvis
- 3) Artery
- 4) Large Intestine
- 5) Disease of lymph glands

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Pituitary gland
- 2) Kidneys and their functions
- 3) Curvature of Vertebrae
- 4) Atlas
- 5) Introduction of digestive system?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Write in detail about digestion & absorption of food in our GIT.
- 2) Explain how yoga therapy can help to cure stress disorder.
- 3) Explain the pulmonary circulation
- 4) Explain the CT Thorax from T1-T12 with the help of diagram



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ELECTRICITY PHYSICS

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) X-ray tube
- 2) Cathode
- 3) Ionization
- 4) Light intensity
- 5) Energy
- 6) Amphere
- 7) Isotope
- 8) Rectification
- 9) Mass Number
- 10) Capacitor

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Fluorescent screen
- 2) Electric capacitor
- 3) Molecules
- 4) State the Joules Law
- 5) Ohm's law

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Properties of a magnet
- 2) The working of a dynamo
- 3) Write about basic chemicals?
- 4) Define isobar?
- 5) Define Cathode?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Explain the production of electric charges and gases in the discharge tube
- 2) Explain about capacitor
- 3) Describe about the laws of Electromagnetic Induction
- 4) Explain about conductors and insulators



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RADIOLOGY PHYSICS

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Diode
- 2) MAS
- 3) Cathode structure
- 4) Cathode
- 5) Exposure
- 6) Heel effect
- 7) Absorbed dose
- 8) Radiation units
- 9) Anode angle
- 10) Ultra sound

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) MA selector
- 2) REM
- 3) Uses of x-rays
- 4) Hard and soft rays
- 5) Rotating anode

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 6) Stationary grid
- 7) X-ray cassette
- 8) Explain Xero radiography?
- 14) Define cathode structure?
- 15) Write ionizing chambers?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Explain about medical radiation dose and exposure
- 2) Describe the x-ray tube with diagram
- 3) Describe the discovery of X-ray
- 4) Describe about radiation hazards



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TWO YEARS (FIRST YEAR) EXAMINATION

POSITIONING IN RADIOGRAPHY

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Scatter radiation
- 2) Drying
- 3) Barium swallow
- 4) Ring finger lateral
- 5) Projection
- 6) MA
- 7) Inter pupillary line
- 8) Contrast media
- 9) The position of wrist AP
- 10) TM joint lateral view

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Hand AP
- 2) Joint PA
- 3) Cervical vertebrae
- 4) Safety measures while doing X ray for women
- 5) Wrist for scaphoid

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Knee joint AP and lateral view
- 2) Anatomy of heart
- 3) Procedure in CT abdomen
- 4) Types of grids
- 5) Define Barium meal?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Describe about the intravenous Urography and explain the technique & Position
- 2) Describe about the intravenous cerography
- 3) Explain special investigations of internal organs
- 4) Explain IVB?



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TWO YEARS (FIRST YEAR) EXAMINATION

DARK ROOM TECHNIQUES

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) KVP
- 2) Scalpula (LV)
- 3) Inverse square law
- 4) Developer
- 5) Define Base?
- 6) Define safelights?
- 7) Write about the cleaning cassettes?
- 8) Write about the back side of the cassettes?
- 9) Write about contains of the racks?
- 10) Define sharpness?

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Draw the dorsal spline (AP view)
- 2) Left hand oblique view
- 3) Radiography
- 4) Advantages of rinsing
- 5) Write about the various types of automatic processor & explain chemical & replenish system?

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Write about defined affective dose & formula?
- 2) Write about explain chemical system and transport system?
- 3) Write functions of fixer?
- 4) Write any one care of Screen?
- 5) Write introduction of x-ray?

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Describe about automatic processor
- 2) Explain about important radiographic accessories
- 3) Explain about ingredients of fixing agent
- 4) Write about Automatic film processor?



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DIPLOMA IN X-RAY TECHNOLOGY

TWO YEARS (FINAL YEAR) EXAMINATION

SPECIAL INVESTIGATION

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Town view
- 2) Lateral view
- 3) Indications for sialography
- 4) Main factors in radiography
- 5) Gall bladder
- 6) Barium enema
- 7) Sialogram
- 8) Ureters
- 9) Nephrogram
- 10) Contrast media

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Meningioma
- 2) Parathyroid glands
- 3) Left lung
- 4) The structure of IV pyelography with the help of diagram
- 5) OCG

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) MCU
- 2) Reactions of contrast media
- 3) Dental formula
- 4) HSG
- 5) Cystogram

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Write in detail about preparation, indications and procedure of IVP
- 2) Write in detail about preparation indications and procedure of Ba enema
- 3) Explain barium meal whole GI tract barium swallow and barium enema
- 4) Explain about hystercal phonograms (HSG)



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DIPLOMA IN X-RAY TECHNOLOGY

TWO YEARS (FINAL YEAR) EXAMINATION

RADIATION HAZARDS & SAFETY MEASURE

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Radiographic positioning
- 2) Gonad shield
- 3) Rectification
- 4) Radio activity
- 5) TLD
- 6) Film badge
- 7) Fluoroscopy
- 8) Dosimeter
- 9) MPD
- 10) Equivalent dose

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Heat factor in X-ray
- 2) Step wedge method
- 3) Fluoroscopy protection
- 4) Characteristics curve
- 5) Heat and three ways of heat transfer

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Teletherapy
- 2) Film badge
- 3) RAD and REM
- 4) Dose measurement
- 5) Grids

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Explain the ways of heat transfer
- 2) Explain late and immediate effects of radiation
- 3) Explain about measures against radiation hazards
- 4) Describe about radiation protection in fluoroscopic procedure



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TWO YEARS (FINAL YEAR) EXAMINATION

HOSPITAL PRACTICE & PATIENT CARE

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Renal colic
- 2) Sedation
- 3) Hypertonic
- 4) Consent form
- 5) Immobilization
- 6) Care of children
- 7) Reactions of contrast media
- 8) Uses of contrast media
- 9) Contrast administration
- 10) How can we reduce reactions of contrast media

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Patient's preparation for abdomen and pelvis
- 2) Hygiene in X-ray treatment
- 3) X-ray tubes
- 4) Resolution
- 5) HSG

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Positive contrast media
- 2) Preparation of the patient for Barium enema
- 3) Methods of introduction of contrast media
- 4) Sialography
- 5) Barium swallow

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Describe disorders after contrast media
- 2) Explain barium swallow techniques
- 3) Explain about the importance of hygiene in the X-ray department
- 4) Make an analysis on positive and negative contrast media



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RADIOGRAPHIC PHYSICS

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Exposure
- 2) Exposure times
- 3) Circuit breaker
- 4) Cathode rays
- 5) Inverse square law
- 6) Ionization
- 7) Focal spot
- 8) Safe light
- 9) Voltmeter
- 10) Radiation

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) Secondary radiation
- 2) Saturable reactor
- 3) Functions of fixer
- 4) Explain radiation effects
- 5) The body parts of CT scanner used

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Fluoroscopy
- 2) Dental x-ray equipment
- 3) Magnets used in MRI
- 4) Portable unit
- 5) MRI

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Explain the pathology of eye, ear and nose
- 2) Describe in detail about computed tomography
- 3) Explain the modern equipments in radiology
- 4) Describe CT Scan of brain parameters with basic anatomy and pathology



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TWO YEARS (FINAL YEAR) EXAMINATION

ECG & IMAGING SERVICE

TIME: 3 Hours

Marks: 100

I. SHORT ANSWER QUESTIONS: (10X2 = 20)

- 1) Skull bones
- 2) Ultra sound
- 3) Electrocardiogram
- 4) SA node
- 5) ECG of one cardiac cycle
- 6) X-ray room
- 7) Sinus rhythm
- 8) AV node
- 9) Sino atrial block
- 10) Exercise test

2. WRITE SHORT NOTES ON: (5X3 = 15)

- 1) MRI
- 2) Radiographic features of HSG
- 3) Abdomen obstruction
- 4) Liver cirrhosis
- 5) Abnormal electrocardiogram

3. WRITE SHORT NOTES ON: (5X5 = 25)

- 1) Genetic effects of radiation
- 2) Sinus rhythm
- 3) Mammography
- 4) Absorbed dose
- 5) Radioactivity

4. ESSAY QUESTIONS: (4X10 = 40)

- 1) Explain heart with the help of neat diagram
- 2) Describe radiographic appearance of IVU series
- 3) Explain the method to analyze the electrocardiogram
- 4) Describe about subendocardial infarction with the aid of diagram